

Practical ML aka MLOps

Antonio Rueda Toicen
Data Science Retreat
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About me

- Background in startups and academia (computer science and bioengineering)
 - Organizer of the [Berlin Computer Vision Group](#)
 - Arepa-lover (try them, they're *awesome*)



Fig 1: Arepas



Berlin Computer Vision Group

★★★★★ (81) ?

📍 Berlin, Germany

👤 879 members · Public group ⓘ

👤 Organized by **Antonio Rueda Toicen**

 [Edit group info](#)

Share:     

<https://www.meetup.com/Berlin-Computer-Vision-Group/>

Agenda

- MLOps as a survivalist bag of tricks
- Intro to MLOps
 - Experiment tracking
 - Deployment in the MLOps lifecycle
 - Monitoring (tracking drift)
- What is FastAPI?
 - REST APIs vs Serverless solutions
- Routes in FastAPI
- A simple text-only example: *Hola mamá*

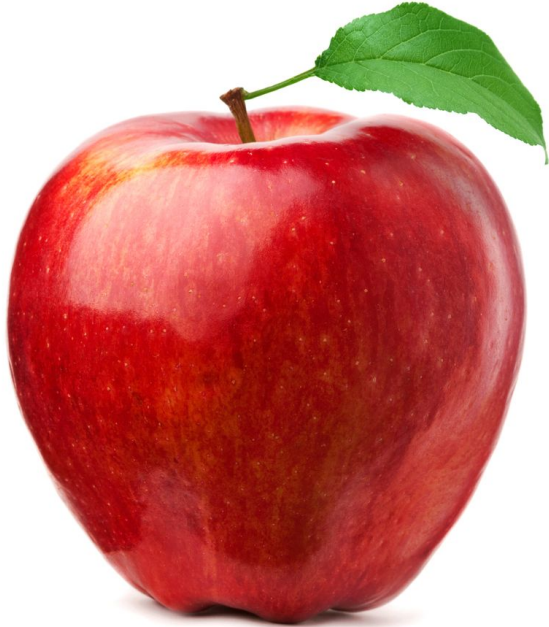
Workflow

1. Train a Resnet18 model on Kaggle notebooks and log all artifacts (trained model and metrics) into wandb
2. Putting this wandb-saved model into a FastAPI endpoint for serving this happens on a Github Codespace
3. Create a Docker image with the FastAPI endpoint
4. Deploy the Docker image with Google Cloud Run

Agenda

- Intro to Google Cloud
- [BigQuery and Window Functions](#)
- Considerations of cloud storage (buckets)
- Weights and Biases
- FastAPI
- Google Cloud Build
- Google Cloud Run

Something to train and deploy



Input

```
{  
  "category": "freshapple",  
  "confidence": 0.9662402272224426  
}
```

Output

[We will be building this model inference endpoint](#)

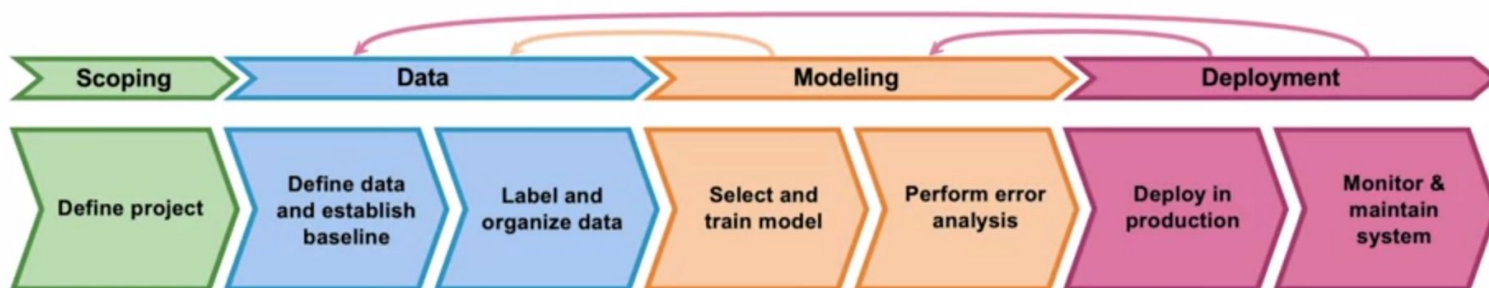
MLOps as a survivalist bag of tricks

- “The business value of an ML model that can only run in a notebook is often 0 EUR” most people in industry today
 - Caveats:
 - Analysis are often necessary and depend on models
 - You don't have to (and must not) deploy everything that you build

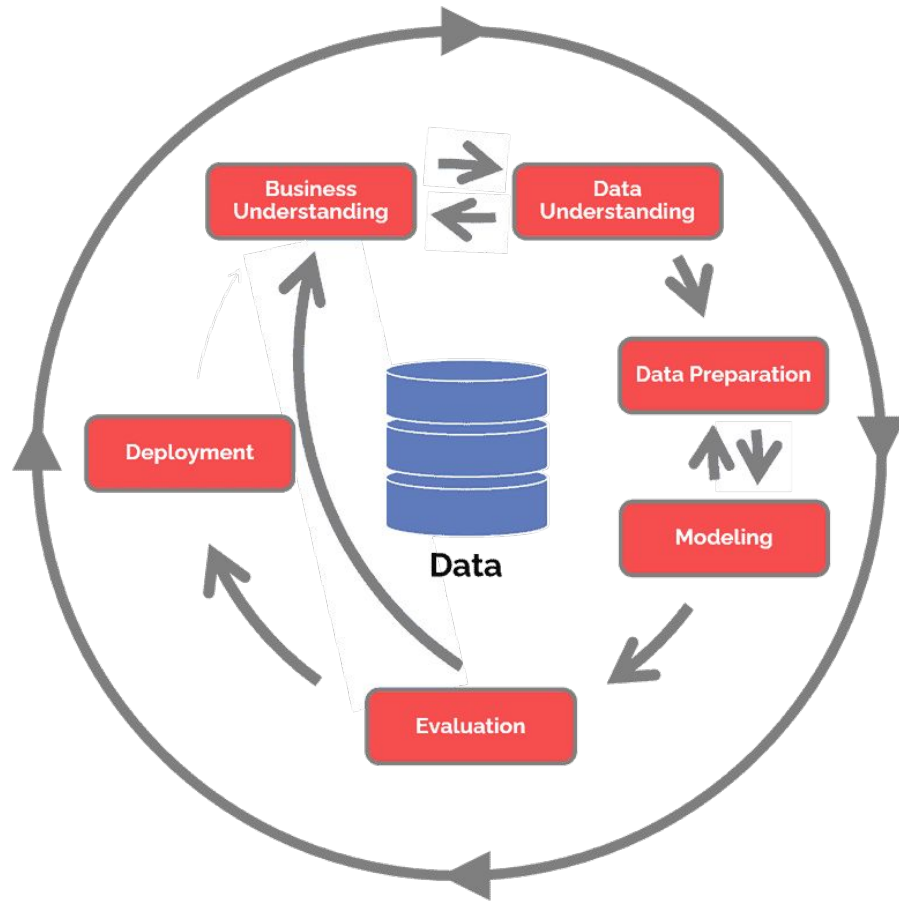
MLOps as a survivalist bag of tricks

- **Experiment tracking**
 - Hyperparameter tuning
 - Data and model version control
- **Reproducibility of runtime environments**
 - Docker
- **Deployment with HTTP endpoints**
 - Monitoring
 - Retraining
- **Use of cloud providers**
 - AWS, Azure, Google Cloud
- **Data pipelines**
 - Airflow, Prefect, Apache Flink

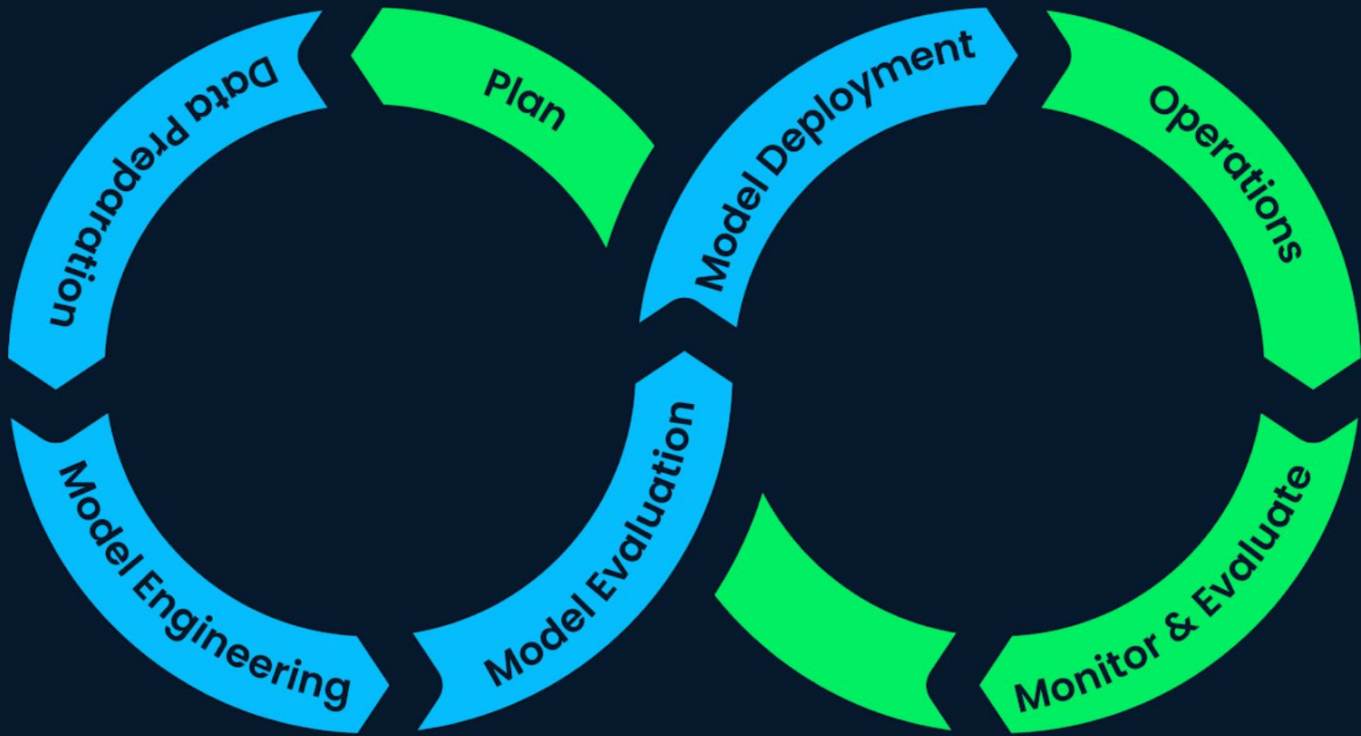
The ML project lifecycle



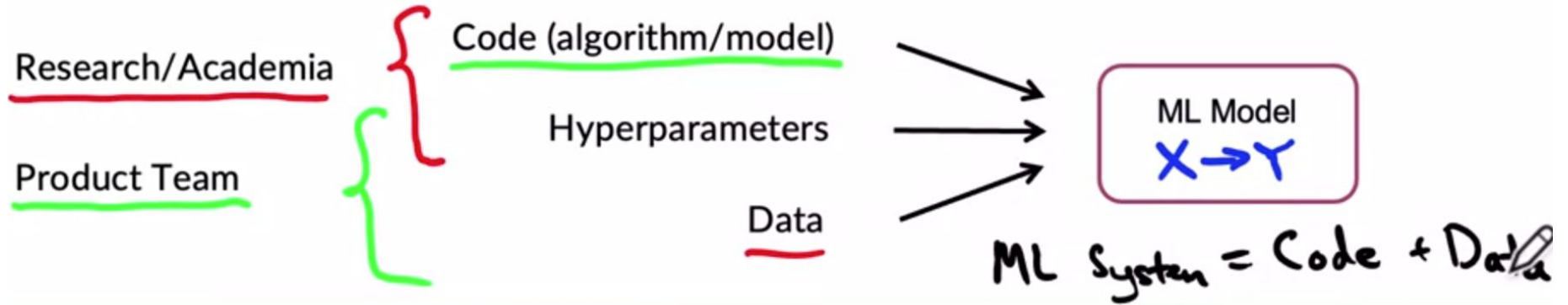
[From Andrew Ng's Intro to ML in Production](#)



[CRISP-DM](#)

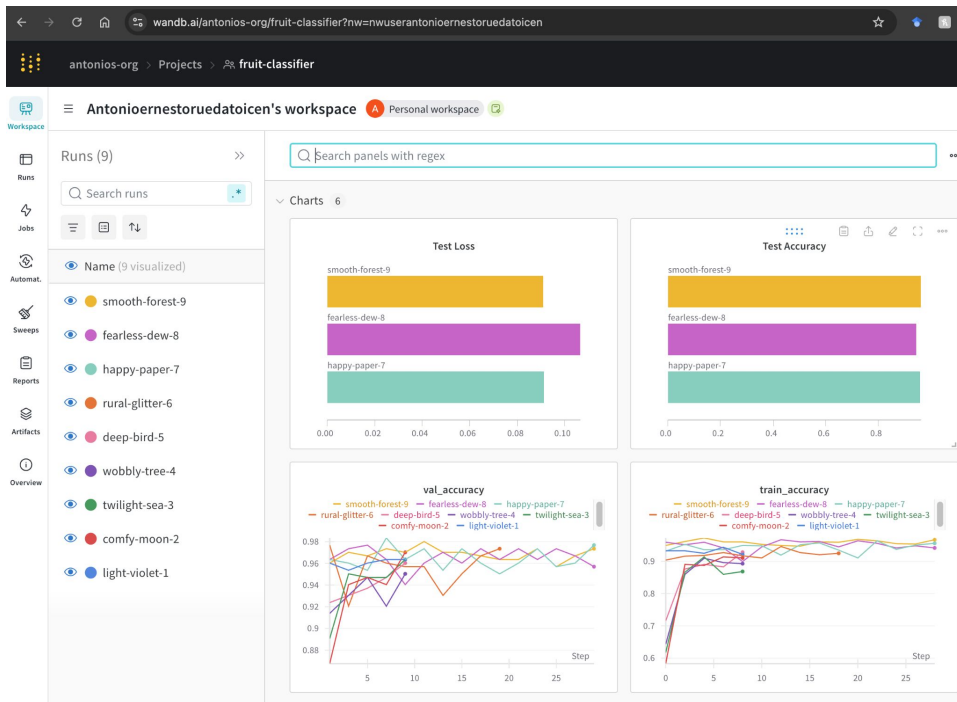


The data driven-approach to ML



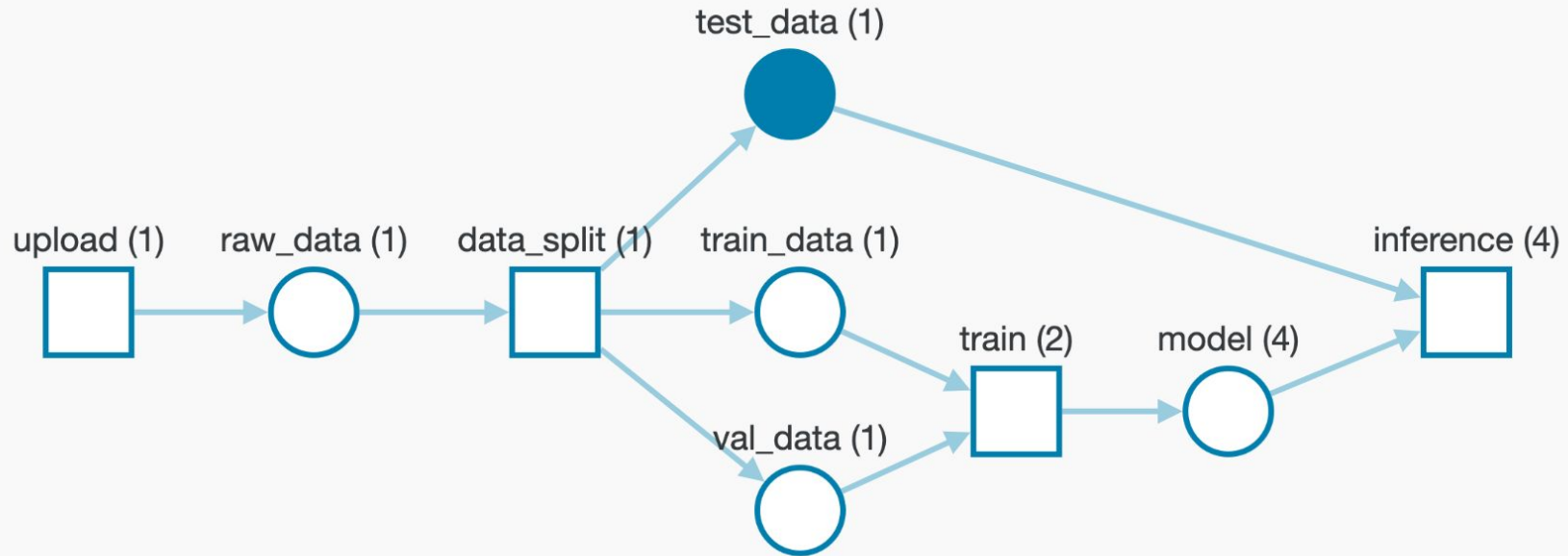
[From Andrew Ng's Intro to ML in Production](#)

Weights and Biases



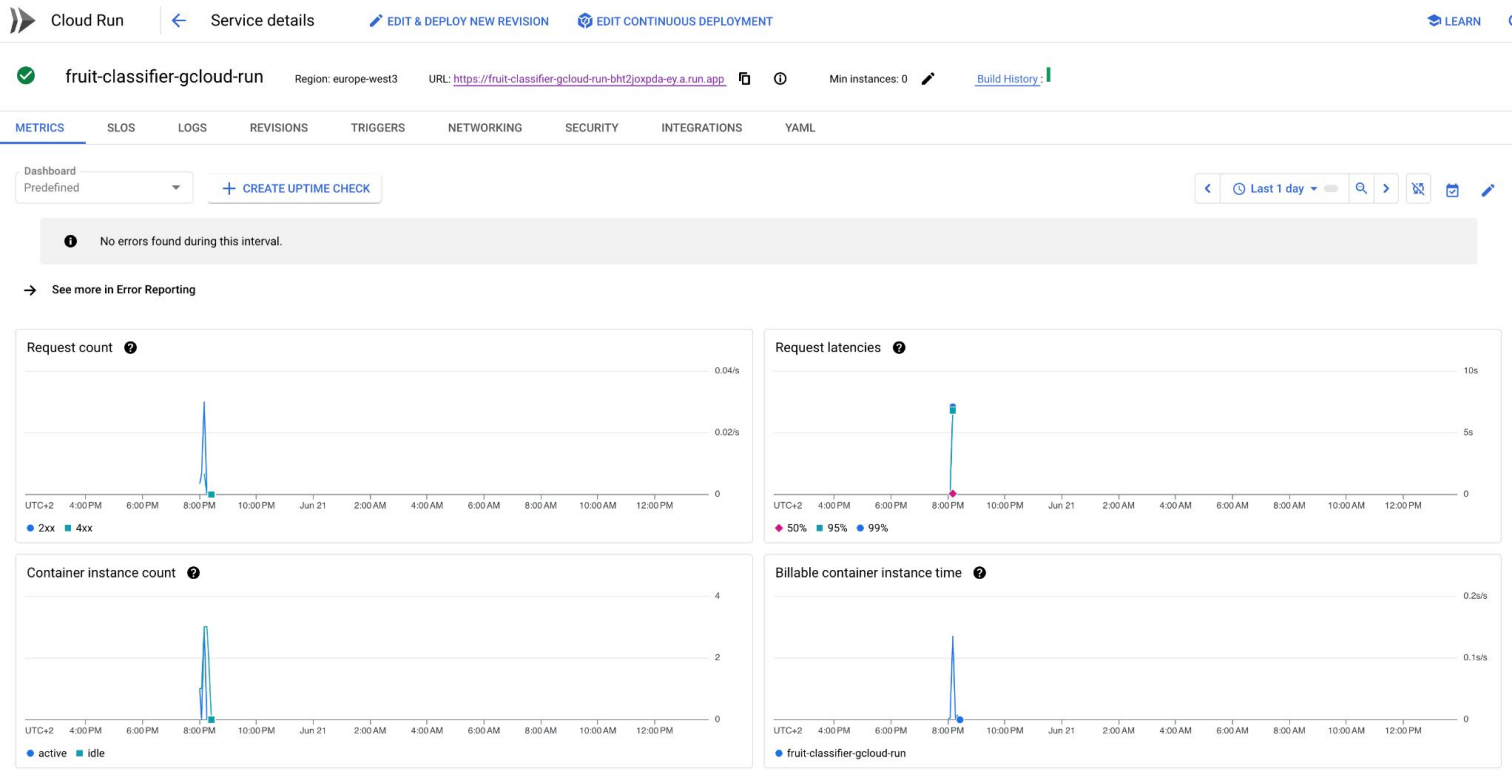
<https://wandb.ai>

Weights and Biases



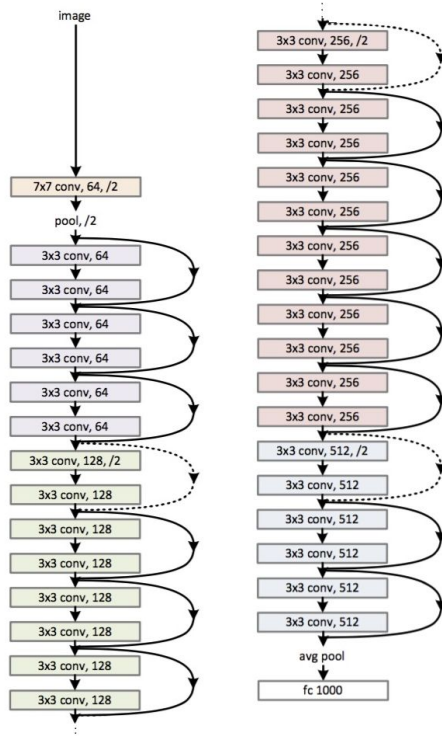
<https://wandb.ai>

From a notebook to an endpoint



What neural network is under the hood

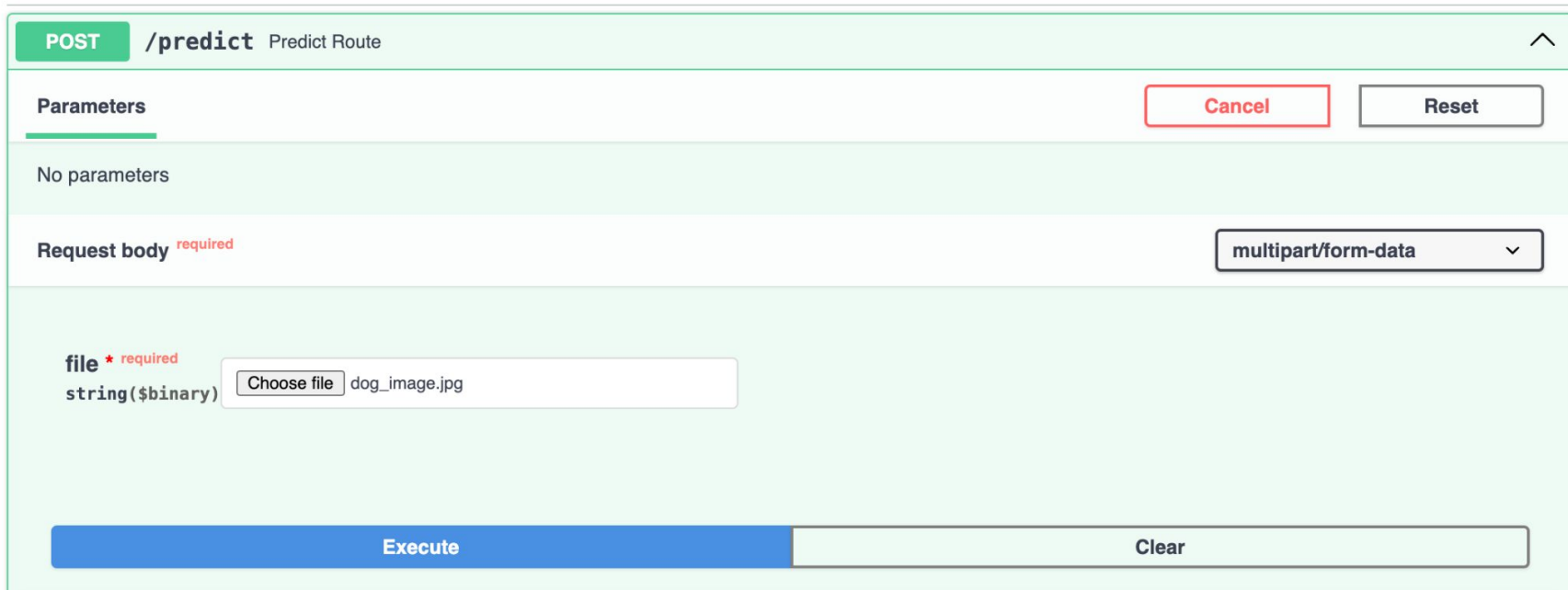
34-layer residual



We will run a Resnet34 network pretrained on the ImageNet dataset (14 million images, 1000 object classes)

What we will be using to get predictions

default



The image shows the SwaggerUI interface for a REST API endpoint. At the top, a green bar displays the HTTP method 'POST' and the endpoint path '/predict' with the description 'Predict Route'. Below this, the 'Parameters' section is empty, showing 'No parameters'. To the right of this section are 'Cancel' and 'Reset' buttons. The 'Request body' section is marked as 'required' and features a dropdown menu set to 'multipart/form-data'. Underneath, a file upload field is shown with the label 'file * required' and 'string(\$binary)'. The file input area contains a 'Choose file' button and the filename 'dog_image.jpg'. At the bottom, there are two large buttons: a blue 'Execute' button and a light green 'Clear' button.

POST /predict Predict Route

Parameters

No parameters

Request body required

multipart/form-data

file * required
string(\$binary)

Choose file dog_image.jpg

Execute Clear

[SwaggerUI](#) comes with FastAPI

The Fruit Classifier Notebook

- [wandb resnet applebananaorange](#)

Create a venv for all local installs

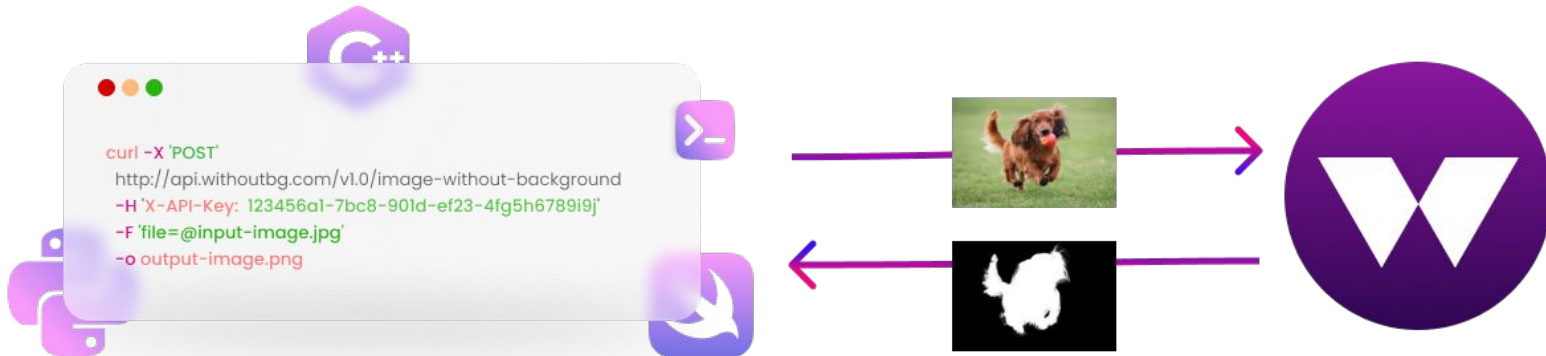
From within the folder where you have created the repo

```
python3 -m venv .venv
```

```
source venv/bin/activate
```

```
pip install -r requirements.txt
```

withoutbg API



FastAPI
ONNX
Docker
DynamoDB
RDS
ECS

What is FastAPI?



- Modern Python web framework for APIs
- Automatic [OpenAPI](#) documentation thanks to [Pydantic](#)
- High performance, faster than most other Python async libraries

<https://fastapi.tiangolo.com/>

The HTTP methods in ML

GET

- Retrieves data from the server
- Example: check if a model is available

POST (not idempotent) or PUT (idempotent)

- Send a data point to get a prediction
- Example: send an image to the endpoint to get its label

Routes in FastAPI

- Routes = endpoints that expose the API
- Each route has a path and an HTTP method (GET, POST, PUT, DELETE)
- Routes define what happens when there is a request to a Uniform Resource Identifier (URI)

Quickstart in FastAPI



```
1 from fastapi import FastAPI
2
3 app = FastAPI()
4
5 @app.get("/")
6 def read_root():
7     return {"message":
8 "Hola"}
```

main.py

Running the simple example

This just takes the first port that it finds available, starting with 8000

uvicorn app.main:app --reload

this specifies the port explicitly, here [main.py](#) is inside the app folder, uvicorn is still the server

fastapi run main.py --port 8080 --reload

Endpoint for the image classification model

```
1 @app.post("/predict/")
2 async def predict_image(file: UploadFile = File(...)):
3     image_data = await file.read()
4     image = Image.open(BytesIO(image_data))
5
6     # Preprocess and predict
7     input_tensor = preprocess_image(image)
8     prediction = predict(input_tensor)
9     prediction = label_mapping[str(prediction)]
10
11     # Now try returning what we have in the notebook (top 5 predictions and their probabilities)
12     return JSONResponse(content={"prediction": prediction})
13
14
```

Validating that the input is an image

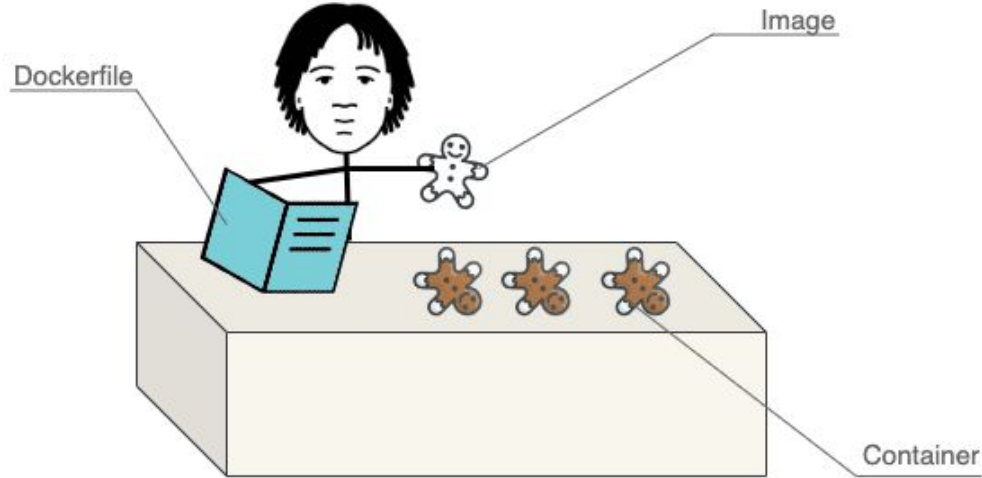


Bringing schema and sanity to your data

[We can validate the input and output schema](#)

Next Step - Creating a Dockerfile

We *could have started with the Dockerfile* (refer to the notes [here](#))



[From testdriven.io](https://testdriven.io)

Environment variables

Artifacts

Registry →

Find matching artifacts

model

resnet18

resnet18

Version 1

Version

Metadata

Usage

Files

Lineage

Version overview

Full Name

antonios-org/banana_apple_orange/resnet18:v1



The full name of our wandb artifact has to be defined as a set of environment variables on our Dockerfile

Environment variables

```
# Set environment variables
ENV WANDB_API_KEY=""
ENV WANDB_ORG="antonios-org"
ENV WANDB_PROJECT="banana_apple_orange"
ENV WANDB_MODEL_NAME="resnet18"
ENV WANDB_MODEL_VERSION="v0"
```

Running the app through Docker

```
docker build -t myfastapiapp .
```

```
docker run -p 8080:8080 myfastapiapp
```

Dockerfiles

Why do we have two?

- Dockerfile
- Convert.dockerfile
- Why one has an extension and the other doesn't?

From an endpoint to a container



[Check out the notes on our docker-compose.yml file](#)

docker-compose.yml

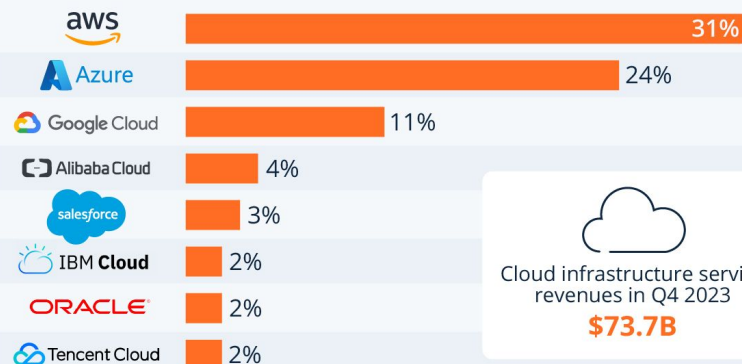
Refer [to the notes](#) to understand the orchestration rules



Biggest Cloud Providers - Q4 2023

Amazon Maintains Cloud Lead as Microsoft Edges Closer

Worldwide market share of leading cloud infrastructure service providers in Q4 2023*




Cloud infrastructure service
revenues in Q4 2023
\$73.7B

* Includes platform as a service (PaaS) and infrastructure as a service (IaaS)
as well as hosted private cloud services

Source: Synergy Research Group



Google Cloud Platform

- Same infrastructure that Google uses for its products
- It offers both servers (aka compute instances that we manage ourselves) and 'serverless' stuff that they manage for us (e.g. Google Cloud Run)
- Divided into regions
 - Choose EU based regions on real projects due to GDPR
- Awesome but **overwhelming**
 - [Many services](#)
 - Google has the habit of rebranding its products mid-development (e.g. Datalab, Vertex AI, Vertex AI Notebooks, etc. have all referred to the same product)
 - AWS is even more complicated, but has less rebranding

Managed vs Unmanaged Services

- “Unmanaged” - we do most of the infra work (e.g. manual Kubernetes)
- “Managed” - Google/Amazon/Microsoft does most of the work (e.g. automated Kubernetes)

Unmanaged Cloud Services

- Provision of raw cloud resources without additional management.
- Control: Full control over the environment for custom configurations.
- Responsibility: Users manage everything (OS, applications, updates, security).
- Benefits: Cheaper; complete customization and control.
- Challenges: Requires significant expertise; time-consuming management; full responsibility for security and compliance.

Managed Cloud Services

- Cloud provider manages infrastructure and software aspects.
- Simplicity: Provider handles management, ensuring smooth operation.
- Expertise: Access to provider's support and management expertise.
- Focus: Users can concentrate on core business or development.
- Benefits: Reduces complexity; expert management; focuses on core activities.
- Challenges: Higher cost; less control over the environment.

‘Serverless’ is unmanaged and on-demand

- Obviously not ‘**serverless**’: there’s always some server serving something something to someone that requests a service
- These are Kubernetes instances that get created on demand to attend requests
- Cheaper to run than using a dedicated server, which would work 24/7

Register now for Google Cloud Next '24 in Las Vegas April 9-11.



Cloud Run

Run applications fast and securely in a fully managed environment

Run frontend and backend services, batch jobs, deploy websites and applications, and queue processing workloads without the need to manage infrastructure.

Go to console

Contact sales

Product highlights

- ↓ The flexibility of containers with the simplicity of serverless
- ↓ Start with source code and run your app in 20+ regions at once
- ↓ Only pay when your code is running
- ↓ Deploy and build web apps and websites



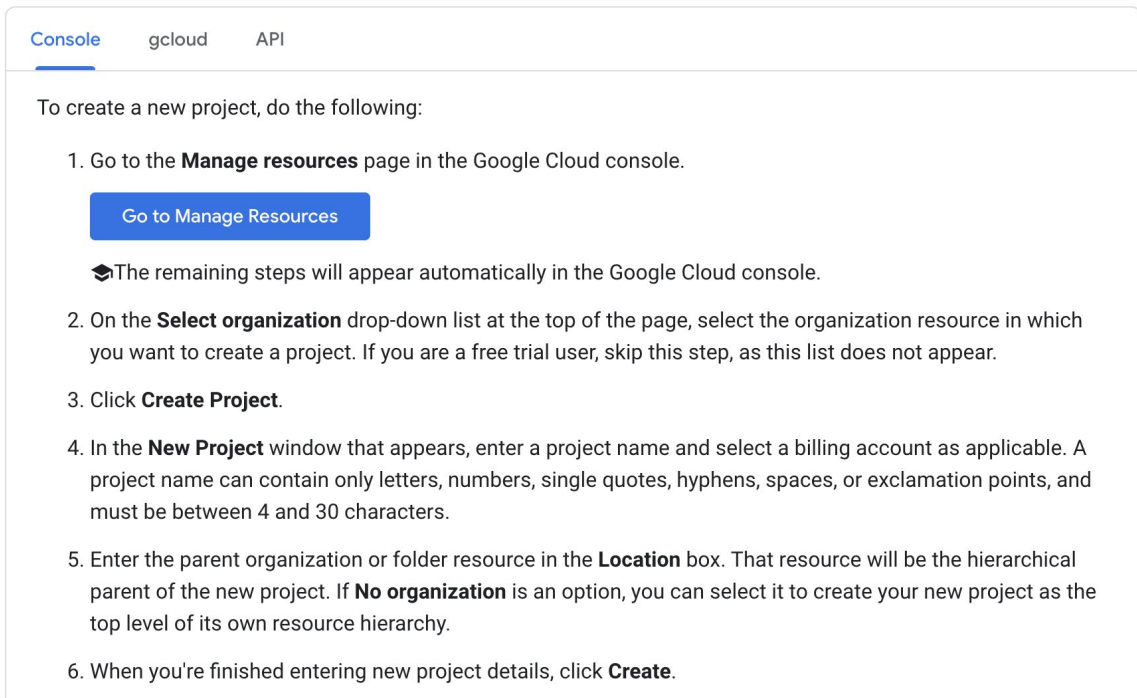
Cloud Run in a minute

1:32

<https://cloud.google.com/run?hl=en>

Creating Projects on Google Cloud

[Creating and managing projects | Resource Manager Documentation | Google Cloud](#)



The screenshot shows the Google Cloud console interface. At the top, there are tabs for 'Console', 'gcloud', and 'API'. The 'Console' tab is selected and underlined. Below the tabs, the text reads: 'To create a new project, do the following:'. This is followed by a numbered list of six steps. Step 1 is 'Go to the **Manage resources** page in the Google Cloud console.' Below this step is a blue button with the text 'Go to Manage Resources'. Step 2 is 'On the **Select organization** drop-down list at the top of the page, select the organization resource in which you want to create a project. If you are a free trial user, skip this step, as this list does not appear.' Step 3 is 'Click **Create Project**.' Step 4 is 'In the **New Project** window that appears, enter a project name and select a billing account as applicable. A project name can contain only letters, numbers, single quotes, hyphens, spaces, or exclamation points, and must be between 4 and 30 characters.' Step 5 is 'Enter the parent organization or folder resource in the **Location** box. That resource will be the hierarchical parent of the new project. If **No organization** is an option, you can select it to create your new project as the top level of its own resource hierarchy.' Step 6 is 'When you're finished entering new project details, click **Create**.'

Console gcloud API

To create a new project, do the following:

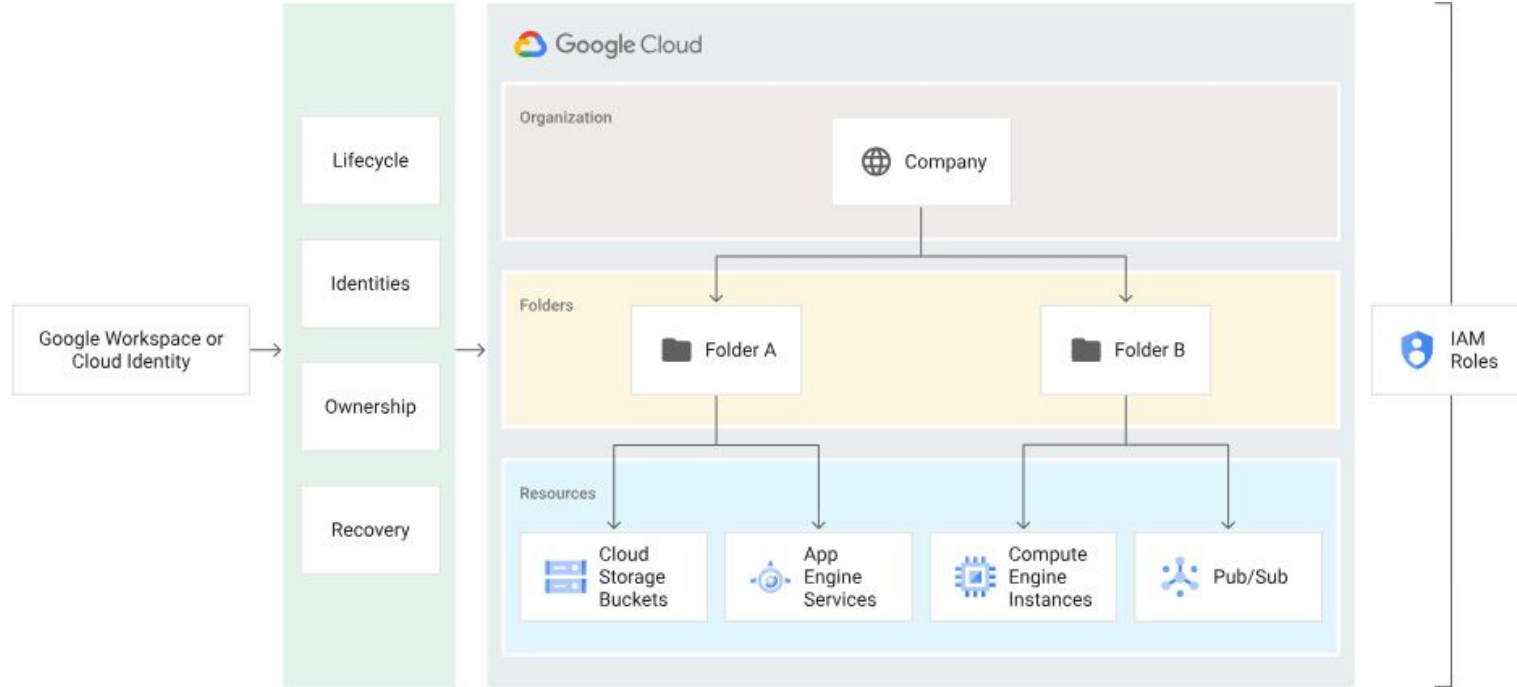
1. Go to the **Manage resources** page in the Google Cloud console.

[Go to Manage Resources](#)

📖 The remaining steps will appear automatically in the Google Cloud console.

2. On the **Select organization** drop-down list at the top of the page, select the organization resource in which you want to create a project. If you are a free trial user, skip this step, as this list does not appear.
3. Click **Create Project**.
4. In the **New Project** window that appears, enter a project name and select a billing account as applicable. A project name can contain only letters, numbers, single quotes, hyphens, spaces, or exclamation points, and must be between 4 and 30 characters.
5. Enter the parent organization or folder resource in the **Location** box. That resource will be the hierarchical parent of the new project. If **No organization** is an option, you can select it to create your new project as the top level of its own resource hierarchy.
6. When you're finished entering new project details, click **Create**.

Identity and Access Management (IAM) Roles



Organization of IAM Roles

Resources

- [Tracking multiple iterations of a dataset in weights and biases](#)
- [Experiment Tracking with Weights and Biases](#)
- [MLOps Courses on Weights and Biases](#)
- [How to create a great dev environment with Docker](#)
- [GitHub - DataTalksClub/mlops-zoomcamp: Free MLOps course from DataTalks.Club](#)